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United States Patent Application Publication

20250279603

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Okada; Yusuke et al.

Connector Combined Body

Abstract

A connector combined body includes a cable side connector and a board side connector electrically connected and mated with the cable side connector. The cable side connector includes a main body part, a movable part connected to the main body part, and a first compound contact disposed in the main body part. The movable part is movable in one direction into the main body part and has a retaining portion in which a cable is inserted and retained. The cable includes a plurality of conductors, a plurality of insulators each separately surrounding one of the conductors, and a sheath surrounding the insulators. A first end side of the first compound contact is an insulation displacement contact. A movement of the movable part in the one direction pushes the insulation displacement contact into contact with the conductors of the cable.

Inventors: Okada; Yusuke (Kawasaki-shi, JP), Takemasa; Eiichiro (Kawasaki-shi, JP), Kato; Hirokazu (Kawasaki-shi, JP)

Applicant: TE Connectivity Japan G. K. (Kawasaki-shi, JP)

Family ID: 96738176

Assignee: TE Connectivity Japan G. K. (Kawasaki-shi, JP)

Appl. No.: 19/063589

Filed: February 26, 2025

Foreign Application Priority Data

JP	2024-030320	Feb. 29, 2024
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Publication Classification

Int. Cl.: H01R12/75 (20110101); H01R4/2429 (20180101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of the filing date under 35 U.S.C. § 119(a)-(d) of Japanese Patent Application No. JP2024-030320, filed on Feb. 29, 2024.

FIELD OF THE INVENTION

[0002] The present disclosure relates to a connector and, more particularly, to a connector combined body.

BACKGROUND OF THE INVENTION

[0003] Conventionally, a crossover wiring method has been employed to make an electrical connection between LED lighting units placed at predetermined intervals. In this method, each LED lighting unit is configured such that a power source or electric input side cable and a power source or electric output side cable, a terminal block, a power source module, and each LED are electrically connected to one another in this order. LED lighting units that are adjacent to each other are connected to each other by a cable. This interconnecting cable can function as an output side cable in one of the LED lighting units and functions as an input side cable in the other of the LED lighting units. Each of the aforementioned input side and output side cables may include a plurality of (e.g. two or three) conductors, insulators separately surrounding each of the plurality of conductors, and a sheath surrounding en bloc the plurality of conductors equipped with the insulators.

[0004] However, prior to making the electrical connection between the cable and the terminal block, a worker needs to perform the task of stripping off part of the sheath of each cable and part of each insulator. For example, a worker can strip off parts of the sheath and each insulator in each cable, expose each conductor inside, insert each conductor into the terminal block, and make an electrical connection between the cable and the terminal block. The task of stripping off part of the sheath of each cable and part of each insulator and exposing part of each conductor inside may decrease working efficiency.

SUMMARY OF THE INVENTION

[0005] A connector combined body includes a cable side connector and a board side connector electrically connected and mated with the cable side connector. The cable side connector includes a main body part, a movable part connected to the main body part, and a first compound contact disposed in the main body part. The movable part is movable in one direction into the main body part and has a retaining portion in which a cable is inserted and retained. The cable includes a plurality of conductors, a plurality of insulators each separately surrounding one of the conductors, and a sheath surrounding the insulators. A first end side of the first compound contact is an insulation displacement contact. A movement of the movable part in the one direction pushes the insulation displacement contact into contact with the conductors of the cable.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] The invention will now be described by way of example with reference to the accompanying figures, of which:

[0007] FIG. 1 is an isometric view of a connector combined body according to an exemplary embodiment;

[0008] FIG. 2A is an isometric view of a second cable side connector (output side) of the connector combined body of FIG. 1;

[0009] FIG. 2B is an isometric view of a first cable side connector (input side) of the connector combined body of FIG. 1;

[0010] FIG. 2C is a top view of the first cable side connector of FIG. 2B or the second cable side connector of FIG. 2A;

[0011] FIG. 2D is a cross-sectional view of the first cable side connector of FIG. 2B or the second cable side connector of FIG. 2A taken along line A-A of FIG. 2C;

[0012] FIG. 2E is a cross-sectional view of the first cable side connector of FIG. 2B or the second cable side connector of FIG. 2A taken along line B-B of FIG. 2C;

[0013] FIG. 2F is a cross-sectional view of the first cable side connector of FIG. 2B or the second cable side connector of FIG. 2A taken along line C-C of FIG. 2D;

[0014] FIG. 3A is an isometric front view of a board side connector of the connector combined body of FIG. 1;

[0015] FIG. 3B is an isometric rear view of the board side connector of FIG. 3A;

[0016] FIG. 3C is a front view of the board side connector of FIG. 3A;

[0017] FIG. 3D is a cross-sectional view of the board side connector of FIG. 3A taken along line D-D of FIG. 3C;

[0018] FIG. 3E is a cross-sectional view of the board side connector of FIG. 3A taken along line E-E of FIG. 3C;

[0019] FIG. 4A is an isometric view of a prior-to-use state of the second cable side connector of FIG. 2A;

[0020] FIG. 4B is an isometric view of a first usage state of the second cable side connector of FIG. 2A;

[0021] FIG. 4C is an isometric view of a second usage state of the second cable side connector of FIG. 2A;

[0022] FIG. 5A is an isometric view of a first usage state of the connector combined body of FIG. 1;

[0023] FIG. 5B is an isometric view of a second usage state of the connector combined body of FIG. 1; and

[0024] FIG. 5C is an isometric view of a third usage state of the connector combined body of FIG. 1.

DETAILED DESCRIPTION

[0025] A configuration of a connector combined body **500** of the present disclosure is described below with reference to FIGS. 1-5C.

[0026] As shown in FIG. 1, the connector combined body **500** comprises a cable side connector **100** and a board side connector **200** that is electrically connectably mated with the cable side connector **100**. That is, the connector combined body **500** is a structural body constituted by a combination of the cable side connector **100** and the board side connector **200**.

[0027] As shown in FIGS. 1 and 2A-2B, the cable side connector **100** includes two cable side connectors **100**. One of the two cable side connectors **100**, as shown in FIG. 2B, is a first cable side connector **100A** for use in a power source or an electric input. As shown in FIG. 2A, the other of the two cable side connectors **100** is a second cable side connector **100B** which functions as a power source or an electric output.

[0028] As shown in FIGS. 2A-2B, each of the first cable side connector **100A** and the second cable side connector **100B** include a main body part **110** in which a plurality of first compound contacts **130** are placed, and a movable part **120** connected to the main body part **110**. The movable part **120**, as shown in FIGS. 4B-4C, is able to move in one direction into the main body part **110**. The number of first compound contacts **130**, as shown in FIGS. 2B, and 2D-2F, that are placed corresponds to the number of conductors **310** that are provided in a cable **300** described in more

detail below. Each first compound contact **130** can be made from a metallic material such as copper.

[0029] As shown in FIGS. 2E-2F, each of the plurality of first compound contacts **130** has an insulation displacement contact **131** at a first end side thereof, has a contact **132** that is a female contact **132** or a male contact **132** at a second end side thereof, and is configured such that the insulation displacement contact **131** and the female contact **132** or male contact **132** are contiguous to each other. The insulation displacement contact **131** can also be referred to as an “IDC contact”.

[0030] As shown in FIGS. 2A-2B, the main body part **110** has a first area **111** in which the insulation displacement contact **131** is able to be placed and a second area **112** that is contiguous to the first area **111** and in which the female or male contact **132** is able to be placed. As shown in FIG. 2A, the second area **112** has an outer surface provided with a catch part **113** that is able to catch a board side connector **200** as described in more detail below.

[0031] As shown in FIG. 2A, the movable part **120** has a retaining portion **121** in which the cable **300** is able to be inserted and retained, as shown in FIGS. 4B-4C. For example, the movable part **120** is able to be moved in one direction into the main body part **110**, from the position shown in FIG. 4B to the position shown in FIG. 4C, by a worker gripping a bottom surface of the main body part **110** and a top surface of the movable part **120** with a tool such as pliers or pincers.

[0032] Because of such a configuration, based on part of the movable part **120** engaging with the main body part **110** prior to insertion of the cable **300**, a large portion of the movable part **120** is located further outward than the main body part **110**. Meanwhile, the movable part **120** can be moved in one direction into the main body part **110** for insulation displacement connection by the insulation displacement contact **131** of the main body part **110** after the insertion of the cable **300**.

[0033] The cable **300**, as shown in FIG. 1, includes a plurality of conductors **310**, insulators **320** separately surrounding each of the plurality of conductors **310**, and a sheath **330** surrounding the plurality of conductors **310** equipped with the insulators **320**.

[0034] For example, the conductors **310** can be made from aluminum, copper, or other substances. The insulators **320** and the sheath **330** can be made from insulating resin. Without limiting the present embodiment, the insulators **320** and the sheath **330** can contain, as one example, at least one type of resin material selected from the group consisting of vinyl resin, phenol resin, epoxy resin, silicone resin, and unsaturated polyester resin. In one example, the cable **300** can be a VVF cable whose insulators **320** and sheath **330** can be made from insulating vinyl resin.

[0035] The sheath **330** can surround en bloc the plurality of conductors **310** equipped with the insulators **320**. As shown in FIG. 1, the plurality of conductors **310** comprise three conductors **310** one of which functions as a grounding conductor and the other two of which function as conductors **310** for use in power source input or power source output to the power source module.

[0036] Movement of the movable part **120** in one direction, as shown in FIGS. 4B and 4C, causes the insulation displacement contact **131** to be pushed into the cable **300** so that each conductor **310** of the cable **300** and each insulation displacement contact **131** contact each other.

[0037] The insulation displacement contact **131**, as shown in FIG. 2E, includes a first contact pin **131a** and a second contact pin **131b** that face each other so as to form a groove portion **131c**. Each contact pin **131a**, **131b** can be configured to be pushed into the cable **300**.

[0038] Specifically, each contact pin **131a**, **131b** is configured such that the width of an upper end side of the groove portion **131c** (i.e. the clearance between a distal end side of the first contact pin **131a** and a distal end side of the second contact pin **131b**) is slightly larger than the diameter of the conductor **310** of the cable **300**, thereby making it easy to guide the conductor **310** into the groove portion **131c**.

[0039] However, the width of a main area of the groove portion **131c** other than the upper end side (i.e. the clearance between a main portion of the first contact pin **131a** other than the distal end side and a main portion of the second contact pin **131b** other than the distal end side) is slightly smaller than the diameter of the conductor **310** of the cable **300** such that both contact pins **131a**, **131b**

pinch and surely make contact with the conductor **310**.

[0040] With such a configuration, each contact pin **131a,131b**, as shown in FIG. 2E, is configured to stick into the insulators **320** and the sheath **330** while not sticking into the conductors **310** of the cable **300**.

[0041] According to such a configuration, the worker does not need to perform the task of stripping off part of the sheath **330** of each cable **300** and part of each insulator **320** and exposing part of each conductor **310** inside. This makes it possible to bring about an improvement in working efficiency in the connection of cables for use in power sources or electric inputs and outputs.

[0042] The main body part **110** can be made from glass-containing polyamide or other substances to ensure strength against an external force. Additionally, the movable part **120** can be made from transparent resin such as polycarbonate or acrylic or other to make it easy to confirm whether each contact pin **131a,131b** has stuck into the sheath **330**.

[0043] As shown in FIGS. 3A and 5A, the single board side connector **200** has a main body part **230** including a first internal space **210A** and a second internal space **210B** equipped with openings that are able to receive the first cable side connector **100A** and the second cable side connector **100B**, respectively. The main body part **230** of the board side connector **200** can be made from glass-containing polyamide or other substances to ensure strength against an external force and to ensure thermal resistance during soldering to a board (e.g. a substrate in the power source module).

[0044] As shown in FIGS. 5A and 5B, the first cable side connector **100A** is inserted into the first internal space **210A** equipped with an opening, and the second cable side connector **100B** is inserted into the second internal space **210B** equipped with an opening. During insertion, the catch part **113** provided on the outer surface of the main body part **110** of the cable side connectors **100A, 100B** catches the board side connector **200**, whereby the state of mating of the first and second cable side connectors **100A** and **100B** with the board side connector **200** can be maintained.

[0045] As shown in FIGS. 3A and 5A, an inside of the board side connector **200** has a second compound contact **220** for contact with the board (e.g. the substrate in the power source module) for a power source or electric input and output. The number of second compound contacts **220** corresponds to the number of conductors **310** that are provided in the cable **300**. The second compound contact **220** can be made from a metallic material such as copper.

[0046] As shown in FIGS. 3A, 3D, and 5A, a first side end of the second compound contact **220** is a contact **221** (**221A,221B**). The contact **221** can be a male contact **221A** or a female contact **221B** that is able to be mated with the female or male contact **132** provided in the first cable side connector **100A** and/or that is able to be mated with the female or male contact **132** provided in the second cable side connector **100B**. As shown in FIGS. 3A, 3D, and 5A, a contact **221A** can be placed in the first internal space **210A**, and a contact **221B** can be placed in the second internal space **210B**. However, as shown in FIG. 3B, a second end side of the second compound contact **220** can be a board connecting contact **222** that is able to be electrically connected to the board.

[0047] In the present embodiment, the contact **221A** placed in the first internal space **210A** and the contact **221B** placed in the second internal space **210B** can be contiguous to each other in the board side connector **200**. Without limiting the present embodiment, in an example, the contact **221A** placed in the first internal space **210A** and the contact **221B** placed in the second internal space **210B** may be separate from each other inside the board side connector **200** and may be contiguous to each other, for example, via a copper foil wire of the board outside the board side connector **200**.

[0048] The foregoing configuration makes it possible to electrically connect the cable side connector **100** to the board side connector **200** by mating of the contacts of each connector **100,200**. This makes it possible to make a suitable electrical connection premised on power source input (or power supply) and power source output between the cable **300** having a plurality of conductors **310** and the power source module via the connector combined body **500**. That is, the connector combined body **500** is interposed between the cable **300** and the power source module so that the cable **300** and the power source module are able to be electrically connected to each other.

[0049] Using the connector combined body **500** of the present disclosure makes it possible to constitute an LED lighting unit by electrically connecting the cable **300**, the connector combined body **500**, the power source module, and an LED to one another in this order. Additionally, doing so makes it possible to place a plurality of the LED lighting units at predetermined spacings and electrically connecting adjacent LED lighting units to each other.

[0050] A method for using the cable side connector **100** of the connector combined body **500** is now described with reference to FIGS. **4A-4C**.

[0051] First, as shown in FIG. **4A**, in a prior-to-use state of the cable side connector **100**, cable side connectors **100** are prepared for use as the first cable side connector **100A** and the second cable side connector **100B**. Although FIGS. **4A-4C** show an example in which the second cable side connector **100B** is used, similar procedures are done for the first cable side connector **100A**.

[0052] Next, as shown in FIG. **4B**, in a first usage state of the cable side connector **100**, the cable **300** is inserted into the cable retaining portion **121** of the movable part **120** of the first cable side connector **100A** and the second cable side connector **100B**. Next, in a second usage state of the cable side connector **100**, the movable part **120** is moved, from the position shown in FIG. **4B** to the position shown in FIG. **4C**, in one direction into the main body part **120** with a tool such as pliers or pincers. Such movement of the movable part **120** in one direction causes the insulation displacement contact **131** to be pushed into the cable **300** so that each conductor **310** of the cable **300** and each insulation displacement contact **131** make contact with each other.

[0053] According to such a configuration, the worker does not need to perform the task of stripping off part of the sheath **330** of each cable **300** and part of each insulator **320** to expose part of each conductor **310** inside. This makes it possible to bring about an improvement in working efficiency in the connection of cables for use in power sources or electric inputs and outputs. Thus, contact is made between each insulation displacement contact **131** of each first cable side connector **100A** and second cable side connector **100B** and each conductor **310** of the cable **300**.

[0054] A method for using the connector combined body **500** after the first cable side connector **100A** and the second cable side connector **100B** are in the second usage state of the cable side connector **100** shown in FIG. **4C**, is now described with reference to FIGS. **5A-5C**.

[0055] First, as shown in FIG. **5A**, in a first usage state of the connector combined body **500**, the preparation is done for inserting the first cable side connector **100A** into the first internal space **210A** of the board side connector **200**. Further, as further shown in FIG. **5A**, preparation is also done for inserting the second cable side connector **100B** into the second internal space **210B** of the board side connector **200**.

[0056] Next, as shown in FIG. **5B**, in a second usage state of the connector combined body **500**, the insertion of the first cable side connector **100A** into the first internal space **210A** of the board side connector **200** is started. Further, as further shown in FIG. **5B**, the insertion of the second cable side connector **100B** into the second internal space **210B** of the board side connector **200** is started.

[0057] After that, the insertion is continued so that the insertion of the first cable side connector **100A** into the first internal space **210A** of the board side connector **200** is completed, as shown in a third usage state of the connector combined body **500** shown in FIG. **5C**. Further, the insertion of the second cable side connector **100B** into the second internal space **210B** of the board side connector **200** is completed, as further shown in FIG. **5C**.

[0058] A connector combined body assembly is the connection, as shown in FIG. **5C**, of the first cable side connector **100A**, connected to the cable **300**, into the first internal space **210A** of the board side connector **200** and the second cable side connector **100B**, connected to the cable **300**, into the second internal space **210B** of the board side connector **200**.

[0059] The foregoing makes it possible to electrically connect the cable side connector **100** to the board side connector **200** by mating the contacts of each connector **100,200** to each other. This makes it possible to make a suitable electrical connection premised on power source input (or power supply) and power source output between the cable **300** having a plurality of conductors **310**

and the power source module via the connector combined body 500 of the present disclosure. [0060] Although embodiments of the present disclosure have been described above, the present disclosure is not limited to these embodiments. Various changes based on the knowledge of persons skilled in the art, such as a combination of the forgoing configurations, can be made without departing from the scope of the claims.

Claims

1. A connector combined body, comprising: a cable side connector including a main body part, a movable part connected to the main body part, and a first compound contact disposed in the main body part, the movable part is movable in one direction into the main body part and has a retaining portion in which a cable is inserted and retained, the cable includes a plurality of conductors, a plurality of insulators each separately surrounding one of the conductors, and a sheath surrounding the insulators, a first end side of the first compound contact is an insulation displacement contact, a movement of the movable part in the one direction pushes the insulation displacement contact into contact with the conductors of the cable; and a board side connector electrically connected and mated with the cable side connector.
2. The connector combined body of claim 1, wherein the cable side connector is one of a pair of cable side connectors, one of the pair of cable side connectors is a first cable side connector and the other of the pair of cable side connectors is a second cable side connector, the first cable side connector and the second cable side connector each mate with one board side connector.
3. The connector combined body of claim 1, wherein the board side connector has a second compound contact disposed in the board side connector, a second end side of the first compound contact mates with the second compound contact.
4. The connector combined body of claim 3, wherein the second end side of the first compound contact is a female contact, and the second compound contact has a male contact.
5. The connector combined body of claim 3, wherein the main body part has a first area in which the insulation displacement contact is disposed and a second area that is contiguous to the first area receiving the second compound contact.
6. The connector combined body of claim 5, wherein the second area has an outer surface with a catch part that catches the board side connector.
7. The connector combined body of claim 1, wherein the connector combined body is interposed between the cable and a power source module, the connector combined body electrically connects the cable and the power source module to each other.
8. The connector combined body of claim 2, wherein the board side connector has a board main body with a first internal space and a second internal space separate from the first internal space, each of the first internal space and the second internal space receives one of the pair of cable side connectors.
9. The connector combined body of claim 3, wherein the second end side of the first compound contact is a male contact, and the second compound contact has a female contact.
10. The connector combined body of claim 1, wherein the main body part is made of a glass-containing polyamide.
11. The connector combined body of claim 1, wherein the movable part is made of a transparent resin.
12. A connector combined body assembly, comprising: a cable including a plurality of conductors, a plurality of insulators each separately surrounding one of the conductors, and a sheath surrounding the insulators; and a connector combined body including a cable side connector having a main body part, a movable part connected to the main body part, and a first compound contact disposed in the main body part, the movable part is movable in one direction into the main body part and has a retaining portion in which the cable is inserted and retained, the connector combined

body includes a board side connector electrically connected and mated with the cable side connector, a first end side of the first compound contact is an insulation displacement contact, a movement of the movable part in the one direction pushes the insulation displacement contact into contact with the conductors of the cable.

13. The connector combined body assembly of claim 12, wherein the insulation displacement contact has a first contact pin and a second contact pin that face each other to form a groove portion between the first contact pin and the second contact pin.

14. The connector combined body assembly of claim 13, wherein a width of an upper end side of the groove portion is larger than a diameter of one of the conductors of the cable.

15. The connector combined body assembly of claim 14, wherein a width of a main area of the groove portion, other than the width of the upper end side, is smaller than the diameter of the one of the conductors of the cable.
